

MANISH SONAWANE

Senior AI/ML Engineer

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Portfolio:

<https://manishson.github.io/portfolio/>

EDUCATION

- 2019: B.E.
Dr. BAMU Aurangabad
- 2016: Diploma
MSBTE

TECHNICAL SKILLS

- o Programming: Python, SQL.
- o Machine Learning: NumPy, Pandas, SciPy, Scikit-learn, Seaborn, Matplotlib, Flask,
- o ML/DL Frameworks: TensorFlow, PyTorch
- o Generative AI: GPT-4, Claude, Llama 2, Mistral 7B, BERT, LegalBERT, LangChain, LangGraph, MCP tool development, Prompt Engineering, Function Calling, RAG, Transfer Learning, LSTM, Seq2Seq
- o Natural Language Processing (NLP): NLG techniques, NER, document classification, semantic search
- o Cloud Platforms: Google Cloud Platform (GCP), Amazon Web Services (AWS) (EC2, SageMaker, S3), Microsoft Azure (Cognitive Services, Azure Blob Storage, Azure SQL), Azure Databricks, Pipeline Automation, Lambda, DynamoDB, Amazon Bedrock, CI/CD.
- o Infrastructure as Code
- o Data & Backend: FastAPI, Django REST Framework, MongoDB, Databricks (Spark, Delta Lake, MLflow)
- o Streamlit, Reflex UI

PROFILE SUMMARY

- 5+ years designing, developing, and deploying AI/ML and GenAI systems, with deep expertise in Python, LLM-based architectures, and enterprise-scale AI applications.
- Hands-on experience with large language models (GPT-4, Claude, Llama 2, Mistral 7B), RAG pipelines, prompt engineering, function calling, MCP tool development, and agentic workflows built with LangChain/LangGraph.
- Built and deployed computer vision, NLP, and legal-domain AI systems (document classification, NER, and weapon/face detection), reaching 90-95%+ accuracy in production.
- Experienced in secure, production-grade AI deployment: Docker, Kubernetes, CI/CD, model monitoring/observability (Prometheus), and enterprise microservice integration
- Skilled at building interactive AI-driven applications (Streamlit, Reflex UI) enabling real-time user interaction with LLM-powered workflows.

WORK EXPERIENCE

Apr 2025 – Present: Senior AI/ML Engineer | STL Digital

Key Result Areas:

- Designed and developed **MCP (Model Context Protocol)** tools to enable structured interaction between LLMs and enterprise services.
- Developed interactive user interfaces using **Stream-lit** for rapid prototyping and **Reflex UI** for production-grade web experiences.
- Implemented function-calling and tool orchestration logic using **LangGraph** to convert natural language queries into executable actions.
- Collaborated with product and backend teams to translate business requirements into scalable AI solutions.

Dec 2024 – Apr 2025: AI/ML Developer | Freelance

Key Result Areas:

- Designed and developed computer vision models for weapon detection and face matching.
- Built and optimized a face-matching system leveraging deep learning techniques for identity verification.
- Developed APIs using FastAPI to integrate computer vision models with frontend applications.
- Ensured optimized inference performance by leveraging GPU acceleration and model quantization techniques.

Jun 2024 – Dec 2024: AI/ML Developer | Qodequay Technologies Pvt. Ltd.

Key Result Areas:

- Designed and implemented **Named Entity Recognition (NER)** models for Natural Language Processing (NLP) tailored to the property services domain.
- Conducted comparative analysis of LLM outputs with traditional NLP methods to validate improvements in accuracy and scalability.
- Developed and deployed **computer vision models** for specific applications, e.g., image classification, object detection, and optimizing performance and scalability.
- Built and optimized **aggregation pipelines** in MongoDB for data-intensive applications, integrating seamlessly with **Django Rest Framework** to support high-performance APIs.

SOFT SKILLS

- o Attention to Details
- o Communication
- o Team Management
- o Problem-solving
- o Time Management
- o Leadership
- o Adaptability

Additional Projects and Proof of Concepts (POCs)

- Developed and implemented **object detection** models using techniques like **YOLO** (You Only Look Once) and **Faster R-CNN**, achieving high accuracy in detecting multiple objects in images.
- **Document Digitization with OCR and NLP**: Automating text extraction and classification from scanned property/legal documents using OCR and transformer-based NLP models.
- **An AI-powered legal document classification and summarization system** using **Generative AI**.
- PII Extraction and Redaction System for PDF Documents.
- **AI-Powered Legal Document Classification and Summarization System**

Feb 2021 – Jun 2024: ML Engineer | 47Billion

Key Result Areas:

- Collaborating with cross-functional teams to gather requirements and translate them into scalable ML solutions.
- Conducting thorough data analysis and visualization to extract valuable insights for business stakeholders.
- Performing EDA using Pandas, Seaborn, and Matplotlib to select the best feature set for models with different methods and statistical tests.
- Building models for different classification algorithms such as logistic regression, decision tree, and random forest.

PROJECTS

Project Title: Sentinel Bot – AI-Powered Enterprise Automation Assistant

Responsibilities:

- Designed and developed **Sentinel Bot**, an AI-powered enterprise assistant enabling users to perform complex multi-system operations using natural language.
- Built a **function-calling framework** with JSON schema validation to dynamically select and execute appropriate tools across **50+ enterprise services** (GitHub, Jira, Confluence, databases, monitoring tools).
- Implemented **data normalization and ETL pipelines** to transform heterogeneous API responses into a consistent internal format.
- Built context-aware conversation management using **LangGraph** to support multi-turn interactions and stateful follow-up queries.
- Achieved **~95% intent-to-action accuracy**, handling **thousands of daily requests**.

Project Title: Surveillance System – Weapon Detection and Face Matching

Responsibilities:

- Designed & developed a real-time AI-powered surveillance system for weapon detection and face matching using advanced computer vision techniques.
- Implemented deep learning models with YOLO and Faster R-CNN, achieving high accuracy and real-time object detection for weapon identification.
- Built & optimized a face recognition system leveraging deep learning and feature extraction techniques to enhance identity verification.
- Deployed & scaled the system on **AWS Cloud** (EC2, S3, Lambda) for high-performance inference and real-time processing.
- Developed REST APIs using FastAPI, enabling seamless integration with existing security infrastructures.

Project Title: Property Insights Automator – Named Entity Recognition System for Property Services

Responsibilities:

- Spearheaded the development of the document classification system within the Advanced Legal Document Classification and Summarization System, employing machine learning algorithms to achieve a commendable accuracy rate of 92% in categorizing legal documents based on case types.
- Conducted comprehensive exploratory data analysis (EDA) on a dataset comprising 10,000+ legal documents, discerning crucial patterns and trends that informed feature selection strategies.
- Executed meticulous data cleaning and preprocessing techniques, resulting in a 20% reduction in noise and extraneous information, thereby bolstering the resilience of the training dataset.
- Contributed to the system's success by achieving a notable 90% accuracy in the classification of legal documents, fostering enhanced efficiency and mitigating manual effort within case management responsibilities.
- Deploying FastAPI applications on cloud platforms like Azure and using containerization tools like Docker.